

QIUTING LOH, Ph.D.

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RESEARCH INTERESTS

My research interests span antibody engineering and discovery, tumor cell biology, and molecular diagnostics. I am particularly interested in combining wet-lab antibody engineering with emerging high-throughput and computational approaches to advance therapeutic antibody development.

EDUCATION

B.Sc. (Hons.), Applied Science (Biotechnology) | *Universiti Sains Malaysia*

Penang, Malaysia 2007 - September 2011

- Dissertation research: Isolation, identification and genome analysis of bacteriophage
- Dean's List Award (2011)

M.Sc., Molecular Medicine | *Universiti Sains Malaysia (USM)*

Penang, Malaysia 2012 - November 2015

- Dissertation research: Antibody phage display and immunoassay development
- Recipient of MyMaster Scholarship (2012 - 2015)

Ph.D., Biological Science | *Nara Institute of Science and Technology (NAIST)*

Nara, Japan April 2017 - June 2022

- Dissertation research: Tumor cell biology and cell metabolism
- Recipient of NAIST International Scholarship (2017 - 2022)

RESEARCH AND PROFESSIONAL EXPERIENCE

Research Assistant | *Institute for Research in Molecular Medicine, USM*

Penang, Malaysia June 2011 - August 2012

- Designed and developed a gene assembly method using degenerate oligonucleotides (Published in Analytical Biochemistry, 2012)
- Maintained detailed laboratory records to support publication and reproducibility

Research Assistant | *Institute for Research in Molecular Medicine, USM*

Penang, Malaysia September 2015 - March 2016

- Collaborated with Technical University of Applied Sciences Wildau for screening fluorescent dyes for detection of pyrophosphate in enzymatic assays (Published in Scientific Reports, 2017)
- Designed and developed new vector assembly method for antibody chain shuffling (Published in Biotechniques, 2018)
- Supported administrative tracking, inventory management, and research coordination tasks
- Served as a senior lab member, mentoring and training new lab members on molecular biology techniques and lab protocols

Assistant to Chief Editor | *Malaysian Journal of Microbiology, USM*

Penang, Malaysia May 2016 - March 2017

- Reviewed and evaluated scientific manuscripts for methodological rigor, novelty, and impact across a broad range of microbiology and molecular biology topics
- Managed manuscript tracking systems from submission through peer review to publication
- Coordinated communications between authors, reviewers, and editorial staff

Researcher | *Nara Institute of Science and Technology*

Nara, Japan April 2017 - June 2022

- Led an independent research program on pancreatic cancer cell metabolism, designing and executing experiments across molecular biology, biochemistry, and cell biology

- Devised experimental methods, standardized protocols, and troubleshoot complex technical challenges with minimal supervision
- Managed and analyzed large experimental datasets, applying quantitative methods to interpret results and support publication
- Contributed original research findings to a patent application through precise experimental design and documentation
- Prepared formal scientific reports and manuscripts for peer review; maintained comprehensive laboratory documentation

PUBLICATIONS

Articles in Refereed Journals

1. **Loh Q**, Bahara NHH, Choong YS, Lim TS. Assembly of highly diverse genes using degenerate oligonucleotides by temperature cascade. *Analytical Biochemistry*. 2012;431:54-56.
2. Omar N, **Loh Q**, Tye GJ, Choong YS, Noordin R, Glökler J, Lim TS. Development of an antigen-DNAzyme based probe for a direct antibody-antigen assay using the intrinsic DNAzyme activity of a daunomycin aptamer. *Sensors*. 2013:346-355.
3. **Loh Q**, Omar N, Glökler J, Lim TS. IQPA: Isothermal nucleic acid amplification-based immunoassay using DNAzyme as the reporter system. *Analytical Biochemistry*. 2014;463:67-69.
4. **Loh Q**, Leong SW, Tye GJ, Choong YS, Lim TS. Improved Fab presentation on phage surface with the use of molecular chaperone coplasmid system. *Analytical Biochemistry*. 2015;477:56-61.
5. Fischbach J, **Loh Q**, Bier FF, Lim TS, Frohme M, Glökler J. Alizarin Red S for Online Pyrophosphate Detection Identified by a Rapid Screening Method. *Scientific Reports*. 2017;7:Article 45085.
6. Lai JY, **Loh Q**, Choong YS, Lim TS. Cassette hybridization for vector assembly application in antibody chain shuffling. *BioTechniques*. 2018;65:269-274.

Book Chapters

1. **Loh Q**, Lim TS. DNA G-Quadruplex as a reporter system for sensor development. In: Larramendy M (Ed.), *Nucleic Acids: From Basic Aspects to Laboratory Tools*. InTech; 2016:121-136.
2. Second book chapter contribution during MSc training (details available upon request).

CONFERENCE PRESENTATIONS

1. **Loh Q**, Glöker J, Lim TS. Development of a DNAzyme Reporter System using the Isothermal Quadruplex Priming Amplification for Antibody-Antigen Interaction. Oral Presentation. International Congress on Chemical, Biological and Environmental Sciences. Bali, Indonesia. August 8-10, 2013.
2. **Loh Q**, Zoltán K, Glöker J, Choong YS, Lim TS. Terbium (III)-binding tag as reporter system for antibody-antigen interaction. Poster Presentation. International Conference on Molecular Diagnostics and Biomarker Discovery. Penang, Malaysia. October 23-25, 2013.

PATENT CONTRIBUTION

- Contributed original experimental findings to patent application during doctoral research at Nara Institute of Science and Technology (details available upon request)

AWARDS AND FELLOWSHIPS

- Dean's List Award, Universiti Sains Malaysia (2011)
- MyMaster Scholarship, Ministry of Higher Education Malaysia (2012 - 2015)
- NAIST International Scholarship, Nara Institute of Science and Technology (2017 - 2022)

TECHNICAL SKILLS

Molecular Biology and DNA Nanotechnology

- Molecular cloning, PCR, RT-PCR, DNA analysis, and gene assembly
- Protein expression, purification, and Western blotting

- Antibody phage display manually and automated using Thermo Fisher Scientific Kingfisher and antibody library construction
- Immunoassays including ELISA and DNAzyme-based detection systems
- Fluorescence spectrophotometers

Cancer Biology and Cell Biology

- Cell culture, transfection, and cell-based assays
- Bone marrow transplantation and xenografts (mouse model)
- HEK293T and pancreatic cancer cell line handling and analysis
- Metabolic flux analysis using Agilent Seahorse XF Analyzer and metabolic profiling
- Imaging using confocal microscope
- Flow cytometry and cell sorting

Data Analysis and Documentation

- Bioinformatic data analysis and statistical analysis
- Image processing, analysis and quantification
- GraphPad Prism, ImageJ, PyMol
- Microsoft Office Suite including Word, Excel and PowerPoint

Scientific Communication

- Scientific writing, editing, and peer manuscript evaluation
- Oral and poster presentation at international conferences

EDITORIAL EXPERIENCE

- Assistant to Chief Editor, Malaysian Journal of Microbiology (2016 - 2017)
- Manuscript evaluation, peer review coordination, and publication workflow management

LANGUAGES

- English (Fluent), Malay (Native), Chinese (Conversational), Japanese (Intermediate)

REFERENCES

Available upon request.